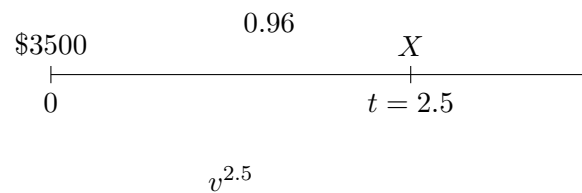


Problem 8.23

At time $t = 0$, Paul deposits \$3500 into a fund crediting interest with an annual discount factor of 0.96. Find the fund value at time 2.5.

Solution

The deposit and the desired fund value can be represented on a timeline:



An annual discount factor of 0.96 means

$$v = 0.96.$$

If X is the fund value at time 2.5, then discounting X back to time 0 gives the original deposit:

$$3500 = X(0.96)^{2.5}.$$

Solving for X ,

$$\begin{aligned} X &= \frac{3500}{(0.96)^{2.5}} \\ &\approx 3876.055. \end{aligned}$$

Therefore, the fund value at time 2.5 is

$$\boxed{\$3,876.06}.$$